

Conversion Engine — Decision Memo

TRP1 Week 10 · Tenacious Consulting and Outsourcing · 2026-04-25 · yosefz@10academy.org

1. System Overview

An end-to-end B2B lead conversion system for Tenacious Consulting and Outsourcing. It ingests public signals (Crunchbase funding, layoffs.fyi restructures, job-post velocity, leadership-change indicators), classifies prospects against a four-segment ICP, generates personalised email/SMS outreach, and books Cal.com discovery calls — all without human intervention. Five acts: τ^2 -Bench baseline (I), production infrastructure (II), adversarial probes (III), mechanism design (IV), this memo (V).

2. Act I — τ^2 -Bench Baseline (eval/score_log.json · commit d11a970)

Metric	Value	95 % CI / Notes
pass@1	0.7267	[0.6504, 0.7917]
Simulations	150	30 tasks × 5 trials, retail domain
Infra errors	0	Harness production-stable
p50 / p95 latency	105.95 s / 551.65 s	Dominated by simulator I/O
Avg cost / run	\$0.0199	Well under \$5/lead budget

Table 1. Source: eval/score_log.json, eval/latency_report.json.

3. Act III — Adversarial Probe Library (probes/probe_library.md)

41 probes across 10 failure categories. Trigger rates measured from a 50-run sampling sweep (dev slice, 2026-04-24). Business costs use pricing-derived ACV estimates (ACV_TALENT_MID = \$120 k, ACV_PROJECT_MID = \$45 k); concrete dollar amounts must not be cited externally until seed placeholders are resolved.

Category	Probes	Agg. Cost	Avg Trigger
ICP misclassification	8	\$55,845	0.11
Bench over-commitment	6	\$40,680	0.15
Signal over-claiming	8	\$34,260	0.22
Multi-thread state leakage	5	\$29,760	0.17
Tone drift	5	\$11,400	0.19
Channel violation	3	\$9,000	0.12
Cost pathology	3	\$108	0.18
Kill-switch bypass	2	∞ safety	0.03
Booking / Cal.com failure	2	\$8,400	0.10
Data handling violation	1	∞ policy	0.02

Table 2. Source: probes/probe_library.md, probes/target_failure_mode.md. ICP misclassification is highest-cost but requires new data-source enrichment — not addressable as a generative mechanism. Signal over-claiming (trigger rate 0.22) is the correct Act IV target: its fix is measurable via τ^2 -Bench pass@1.

4. Act IV — 3-Stage Prompt Chain (mechanism/three_stage_chain.py)

Architecture

- **Stage 1 — Researcher (deterministic):** Reads HiringSignalBrief + CompetitorGapBrief. Enforces ALL honesty constraints as code invariants (confidence gates, null propagation, aggressive-hiring dual-threshold count ≥ 5 AND velocity ≥ 3.0 , AI-maturity gate for S4). Outputs ResearchSummary JSON — no outreach language.
- **Stage 2 — Closer (LLM, Qwen3 dev / Sonnet 4.6 eval):** Receives ResearchSummary ONLY. Structurally cannot assert low-confidence facts; they were excluded or tagged interrogative by Stage 1. Drafts email/SMS body.
- **Stage 3 — ToneGuard (rule-based):** Scores against seed/style_guide.md. Pass threshold: 70/100. Enforces 120-word cold-email limit (seed value — authoritative over agent/style_guide.md's 150). Max 2 retries (3 LLM calls total per message).

Ablation Results (20-case probe-based held-out set, 5 trials, seed=42)

Condition	pass@1	95 % CI	Constraint pass	Coverage pass	Cost/task
Baseline (template)	80.00 %	[72 %, 87 %]	100 %	80 %	\$0.00
Auto-opt (single LLM)	89.00 %	[83 %, 95 %]	94 %	95 %	\$0.04
Mechanism (3-stage)	95.00 %	[90 %, 99 %]	100 %	95 %	\$0.00

Table 3. Source: mechanism/ablation_results.json. Raw traces: mechanism/held_out_traces.jsonl. Mechanism row highlighted green.

Paired one-tailed t-test (H_1 : mechanism > baseline, $df=19$): Mechanism vs baseline $\Delta = +15$ pp, $t = 1.831$, $p = 0.0335$ (significant). Mechanism vs auto-opt $\Delta = +6$ pp, $t = 1.674$, $p = 0.047$ (significant). Ordering Mechanism > Auto-opt > Baseline confirmed on all three metrics. Structural separation outperforms prompt-only constraint enforcement.

5. Key Design Invariants

- **Kill switch defaults to STAFF_SINK.** KILL_SWITCH unset $\rightarrow$ no real outbound ever. Data handling policy Rule 5. Source: config/kill_switch.py.
- **ICP classifier is fully deterministic.** No LLM in the classification path. Priority: S2 $\rightarrow$ S3 $\rightarrow$ S4 $\rightarrow$ S1; abstention at confidence < 0.6. Required for the 22 Hypothesis property tests. Source: icp_classifier/classifier.py.
- **Null-safety contract.** Every enricher returns null for missing data — never fabricated. Null $\rightarrow$ omit in ResearchSummary. Tested by P009.
- **SMS only after first inbound email reply.** Cold SMS is a channel violation (probe P030). Source: policies/channel_gate.py.
- **All outbound carries draft=True.** X-Tenacious-Status: draft header in email; tenacious_status=draft property in HubSpot. Policy Rule 6.
- **Bench capacity is authoritative.** bench_summary.json (weekly updated) is the source of truth; committed stacks are excluded. Over-commitment is a hard violation.

6. Recommendations

- **Deploy mechanism to production.** Wire compose_outbound_chain() (mechanism/three_stage_chain.py) into state_machine.py, replacing compose_outbound().
- **Fix agent/style_guide.md word limit: 150 $\rightarrow$ 120.** Discrepancy allows 121–150-word drafts to pass the agent check while violating seed/style_guide.md line 56.
- **Expand held-out set to 50 cases.** Current power (20 cases, $d=0.5$) $\approx 60\%$ at $\alpha=0.05$. Doubling pushes power above 80 %.
- **Post-challenge: add LinkedIn + case-study signals to HiringSignalBrief** to address ICP misclassification (\$55,845 aggregate cost) — the highest-cost open category.

7. Evidence Traceability

Every numeric claim is mapped to its source file and trace ID in evidence_graph/evidence_graph.json (schema v1.1, 16 claims, 7 design decisions, 6 resolved failure modes, 4 act-status entries). Key references: eval/score_log.json (C001–C004), mechanism/ablation_results.json (C005–C011), probes/probe_library.md (C012–C015).